

Optimization Problem and Objective Function

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1 Introduction

This document presents the optimization problem and objective function for DVR (Distributed Verification Routing), a system that distributes verification work across heterogeneous verifiers across different stages of a Chain-of-Thought reasoning response within a speculative decoding setting.

2 Optimization Problem

2.1 Objective Function

DVR has three things it cares about simultaneously. It must maximise accuracy, ensuring the final answer is correct within a calibrated tolerance of always using the large verifier on every subtask. It must minimise latency, as verification is the bottleneck and reducing verifier calls and using smaller verifiers directly reduces response time. It must minimise energy, as smaller verifiers consume less compute and skipping verification costs nothing.

These three objectives pull against each other. Maximum accuracy wants the large verifier everywhere. Minimum latency and energy want no verifier anywhere. DVR resolves this tension by treating accuracy as a hard constraint and minimising cost subject to that constraint.

Formally, DVR minimises total verification cost across all three subtasks:

$$\min_{v(s)} \sum_{s \in S} C(v(s)) \quad (1)$$

where $S = \{\text{Exploration, Reasoning, Conclusion}\}$, $v(s)$ is the verifier assigned to subtask s , and $C(v(s))$ is the cost of that verifier.

2.2 Decision Variables

DVR has one primary decision variable per subtask and a set of observed variables that inform it.

2.2.1 Primary Decision Variable

For each subtask $s \in S$:

$$v(s) \in \{\text{none}, \text{small}, \text{medium}, \text{large}\} \quad \forall s \in S \quad (2)$$

with cost ordering:

$$C(\text{none}) = 0 < C(\text{small}) < C(\text{medium}) < C(\text{large}) \quad (3)$$

2.2.2 Observed Variables from the Draft Model

The following variables are computed after the draft model completes each subtask.

$p(s)$ is the softmax probability distribution over the vocabulary at each token position in subtask s . It is the raw output of the draft model and the source of both $q(s)$ and $H(s)$.

$q(s)$ is the average peak confidence of the draft model across all token positions in subtask s :

$$q(s) = \frac{1}{\gamma(s)} \sum_{t=1}^{\gamma(s)} \max_w p_t(w) \quad (4)$$

where $p_t(w)$ is the softmax probability assigned to token w at position t , and $\gamma(s)$ is the total number of token positions in subtask s .

$H(s)$ is the average entropy of the softmax distribution across all token positions in subtask s :

$$H(s) = \frac{1}{\gamma(s)} \sum_{t=1}^{\gamma(s)} \left(- \sum_w p_t(w) \log p_t(w) \right) \quad (5)$$

$\gamma(s)$ is the number of tokens generated by the draft model for subtask s , determined by where the boundary token fires.

$\alpha(s)$ is the token acceptance rate for subtask s , recorded after verification completes.

2.2.3 Observed Variables from the System

The following variables are fixed and known before any generation begins.

$C(v)$ is the verification cost per verifier, scaling with parameter count. $L(v)$ is the latency per token for each verifier, where $L(\text{small}) < L(\text{medium}) < L(\text{large})$. L_{draft} is the latency per token of the draft model. $E(v)$ is the energy consumed per token by each verifier, measured in grams CO_2 per token using CodeCarbon [2]. E_{draft} is the energy consumed per token of the draft model. T_{max} is the maximum acceptable total latency per response.

2.2.4 Observed Variables from the Task

s is the stage identity, one of {Exploration, Reasoning, Conclusion}, known before generation begins. Query complexity is estimated from query token length, taking values in {low, medium, high}. Query domain is the subject area of the incoming query, taking values in {mathematics, code generation, common sense reasoning, symbolic reasoning, qu...

2.3 Constraints

2.3.1 Constraint 1: Quality

The combined verifier assignment across all three subtasks must keep accuracy above the large verifier only baseline minus the tolerance δ :

$$\text{Acc}(v(E), v(R), v(C)) \geq \text{Acc}(L, L, L) - \delta \quad (6)$$

where $E = \text{Exploration}$, $R = \text{Reasoning}$, $C = \text{Conclusion}$, $L = \text{large verifier}$, and $\delta = 0.05$ adopted from the Learn Then Test calibration framework from CALM [1]. Thresholds are calibrated on MATH and GSM8K validation sets.

2.3.2 Constraint 2: Latency

The total verification time across all three subtasks must not exceed the maximum acceptable response time:

$$L_{\text{draft}} \times \sum_{s \in S} \gamma(s) + \sum_{s \in S} L(v(s)) \cdot \gamma(s) \leq T_{\text{max}} \quad (7)$$

2.3.3 Constraint 3: Skip Decision

Both $q(s)$ and $H(s)$ jointly determine whether to skip verification for a subtask. Skipping is only appropriate when the draft model was both confident and not uncertain:

$$v(s) = \text{none} \quad \text{if} \quad q(s) \geq \theta_q \quad \text{AND} \quad H(s) \leq \theta_H \quad (8)$$

$$v(s) \in \{\text{small}, \text{medium}, \text{large}\} \quad \text{if} \quad q(s) < \theta_q \quad \text{OR} \quad H(s) > \theta_H \quad (9)$$

where θ_q and θ_H are calibrated jointly on a validation set drawn from MATH and GSM8K using the Learn Then Test framework from CALM [1]. LTT searches over combinations of θ_q and θ_H to find the joint thresholds at which skipping verification still keeps accuracy above the required level. LTT provides a statistical guarantee that these thresholds generalise to unseen test data.

2.4 Full Optimisation Problem

DVR solves the following problem for every incoming query. Each component is explained below before the full problem is stated together.

What DVR is minimising. For every query, DVR assigns one verifier to each of the three subtasks. The total verification cost is the sum of the cost of each assignment. DVR minimises this sum:

$$\min_{v(s)} \sum_{s \in S} C(v(s)) \quad (10)$$

Quality constraint. DVR must not produce answers that are significantly worse than always using the large verifier. The accuracy of the chosen verifier combination across the three subtasks must stay within δ of the large-verifier-only baseline. This threshold is calibrated using the Learn Then Test framework from CALM [1] on MATH and GSM8K validation sets:

$$\text{Acc}(v(E), v(R), v(C)) \geq \text{Acc}(L, L, L) - \delta \quad \delta = 0.05 \quad (11)$$

Latency constraint. The total time spent generating and verifying the response must not exceed the maximum acceptable response time $T_{\max}$. This includes both the draft model’s generation time across all tokens and the verifier’s checking time for each subtask:

$$L_{\text{draft}} \times \sum_{s \in S} \gamma(s) + \sum_{s \in S} L(v(s)) \cdot \gamma(s) \leq T_{\max} \quad (12)$$

Skip decision. After the draft model generates each subtask, the router computes $q(s)$ and $H(s)$ from the draft model’s token probability distributions. If the draft was sufficiently confident, meaning peak confidence is above θ_q and entropy is below θ_H , verification is skipped entirely for that subtask. If either condition fails, a verifier must be assigned:

$$v(s) = \text{none} \quad \text{if} \quad q(s) \geq \theta_q \text{ AND } H(s) \leq \theta_H \quad (13)$$

$$v(s) \in \{\text{small}, \text{medium}, \text{large}\} \quad \text{if} \quad q(s) < \theta_q \text{ OR } H(s) > \theta_H \quad (14)$$

The thresholds θ_q and θ_H are calibrated jointly using LTT [1], providing a statistical guarantee that skipping verification when both conditions are met does not violate the quality constraint on unseen data.

Full problem. Putting everything together, DVR solves:

$$\min_{v(s)} \sum_{s \in S} C(v(s)) \quad (15)$$

subject to:

$$\text{Acc}(v(E), v(R), v(C)) \geq \text{Acc}(L, L, L) - \delta \quad (\text{Quality})$$

$$L_{\text{draft}} \times \sum_{s \in S} \gamma(s) + \sum_{s \in S} L(v(s)) \cdot \gamma(s) \leq T_{\text{max}} \quad (\text{Latency})$$

$$v(s) = \text{none} \quad \text{if} \quad q(s) \geq \theta_q \quad \text{AND} \quad H(s) \leq \theta_H \quad \forall s \in S \quad (\text{Skip})$$

$$v(s) \in \{\text{small}, \text{medium}, \text{large}\} \quad \text{if} \quad q(s) < \theta_q \quad \text{OR} \quad H(s) > \theta_H \quad \forall s \in S \quad (\text{Verify})$$

$$v(s) \in \{\text{none}, \text{small}, \text{medium}, \text{large}\} \quad \forall s \in S \quad (\text{Domain})$$

References

- [1] Schuster, T., et al. *Confident Adaptive Language Modeling (CALM)*. NeurIPS, 2022.
- [2] Wilkins, T., et al. *Chain Routing: Environmentally Aware Routing for Chain-of-Thought Reasoning*. KONVENS, 2025.